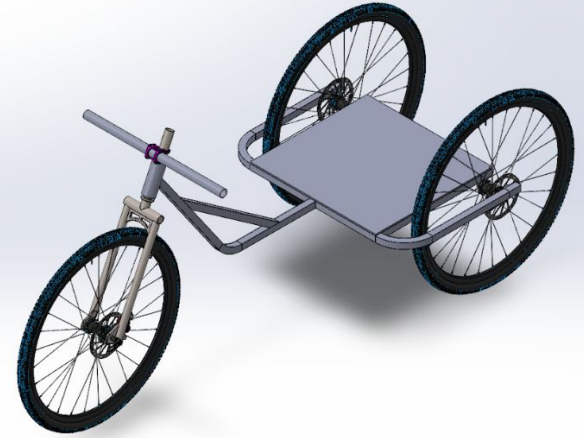




Team members (Right To Left):
Elijah Dunning, Leighanne Bennett,
Ryan Koh, Erik Stern, Karl Kubala,
Rylee Berger

Assembly

Exploded View



Exploded Isometric Assembly View of the 3025-A01-R5 bicycle frame and components.

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	3025-A02	Frame	1
2	3025-B01	Wheels	3
4	3025-B02	Rear Wheel Axles	2
5	3025-C01	Floor Cover	1
6	3025-B03	Brake Caliper	1
7	3025-D01	Stem Coupler	1
8	3025-B04	Front Axle	1
9	3025-D02	Stem	1
11	3025-D03	Handlebars	1
13	91627A15	High Strength Wide Flange Nut	3
14	2299K319	Flat Sprocket	1
15	Drill Plate SubAssembly Version 4	Drill Plate Assembly	1

SPONSOR: Idea Forge
PROJECT: MCEN 3025
TEAM NUMBER: 2Fast2Fail

FILE NAME: 3025-A01-R5

DRAWN BY: Erik Stern

DATE: 2/25/2026

DESCRIPTION: Exploded Isometric Assembly View

3025-A01

1 of 1

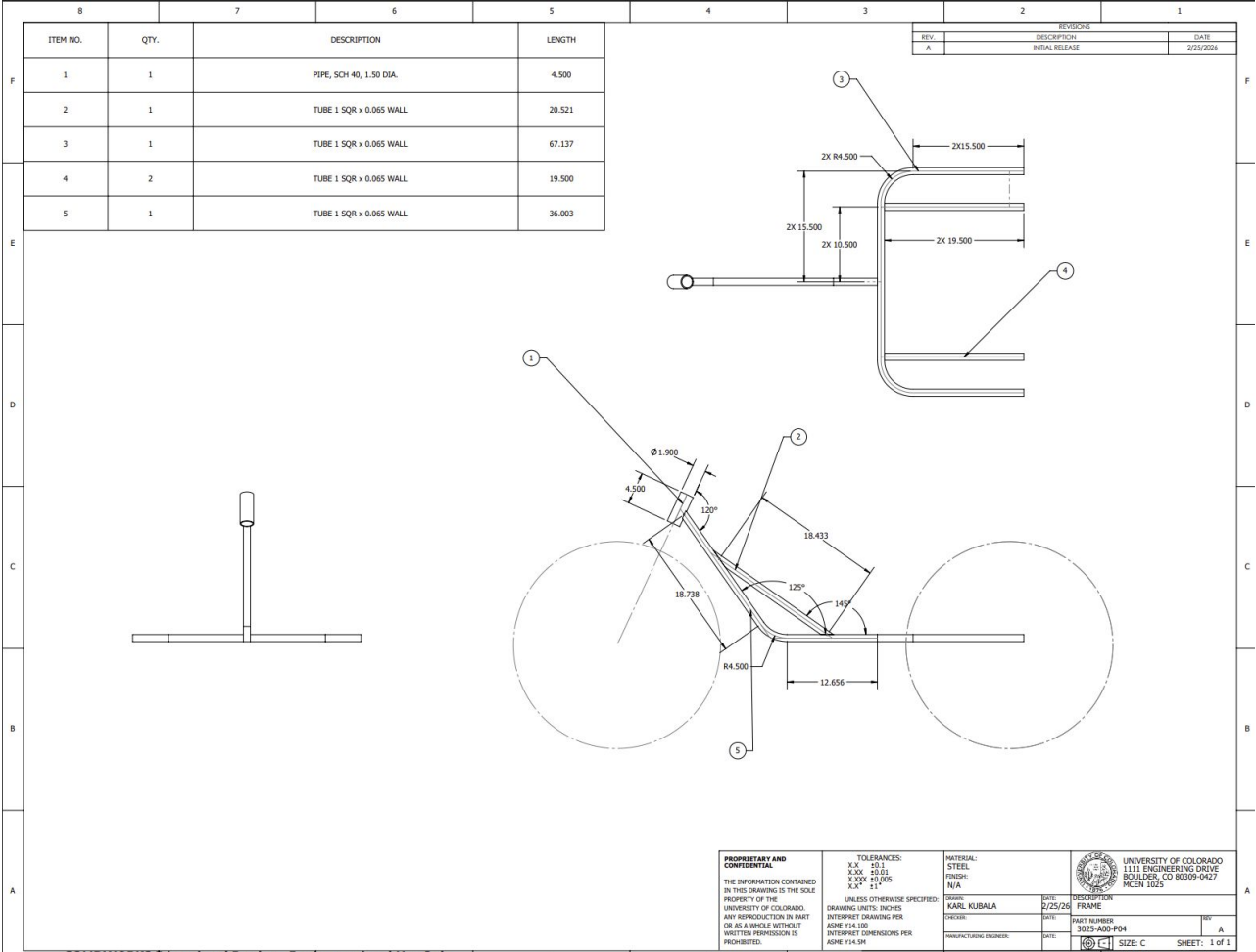
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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	3025-A02	Frame	1
2	3025-B01	Wheels	3
4	3025-B02	Rear Wheel Axles	2
5	3025-C01	Floor Cover	1
6	3025-B03	Brake Caliper	1
7	3025-D01	Stern Coupler	1
8	3025-B04	Front Axle	1
9	3025-D02	Stern	1
11	3025-D03	Handbars	1
13	91627A115	High Strength Wide Flange Nut	3
14	2299K319	Flat Sprocket	1
15	Drill Plate SubAssembly Version 4	Drill Plate Assembly	1

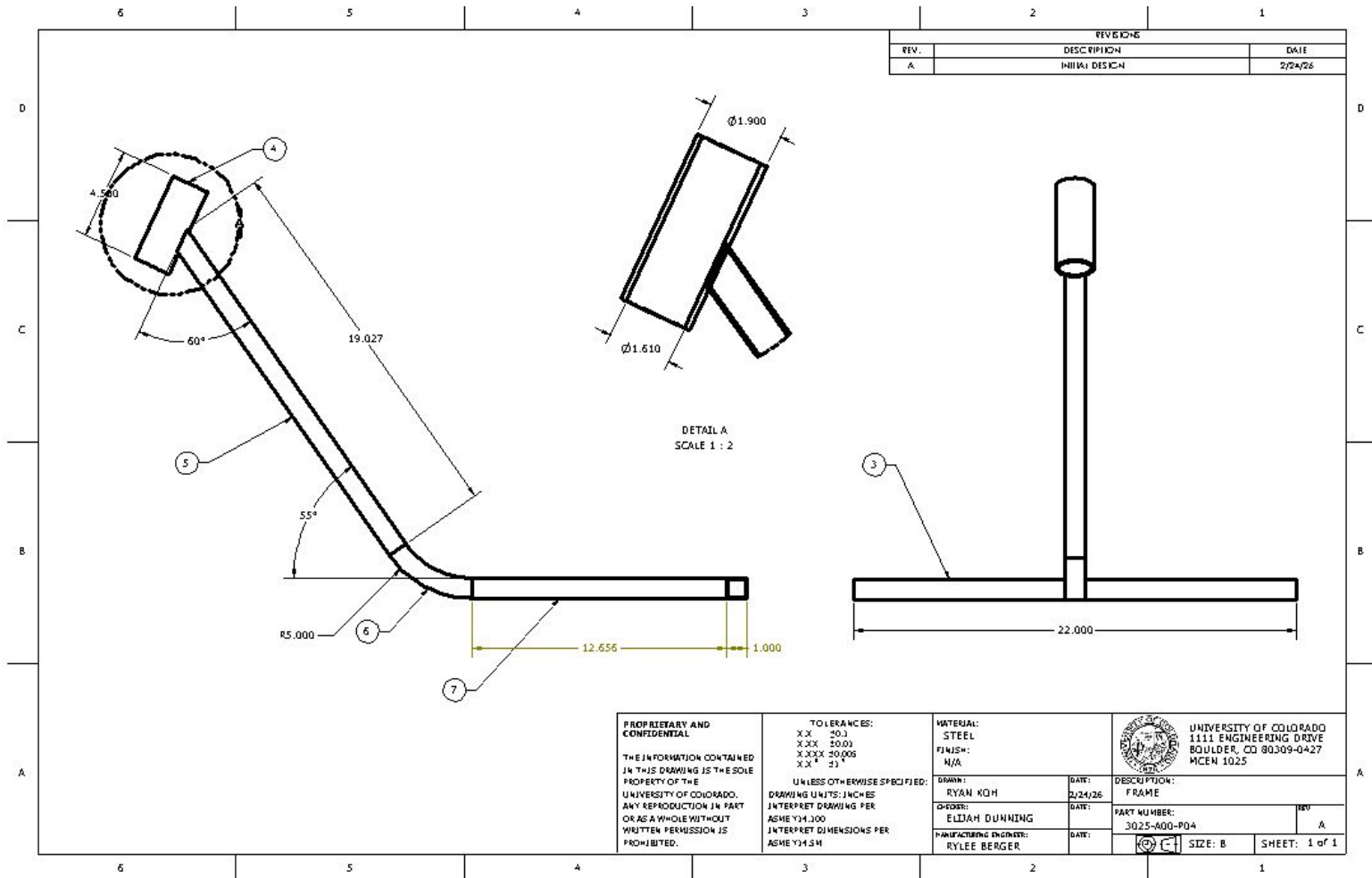
SPONSOR Idea Forge		dc DESIGN CENTER COLORADO A COMMITMENT TO EXCELLENCE	
PROJECT MCIN 3025 2Fest2Fall			
TEAM NUMBER 3025-A01-R5		FILE NAME 3025-A01-R5	
DESIGN BY Erik Stern		DESIGN DATE 2/25/2026	
APPROVAL		CULCATION Explored Isometric Assembly View	
		FR: 3025-A01 REV:  SHEET: 1 of 1	
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Design

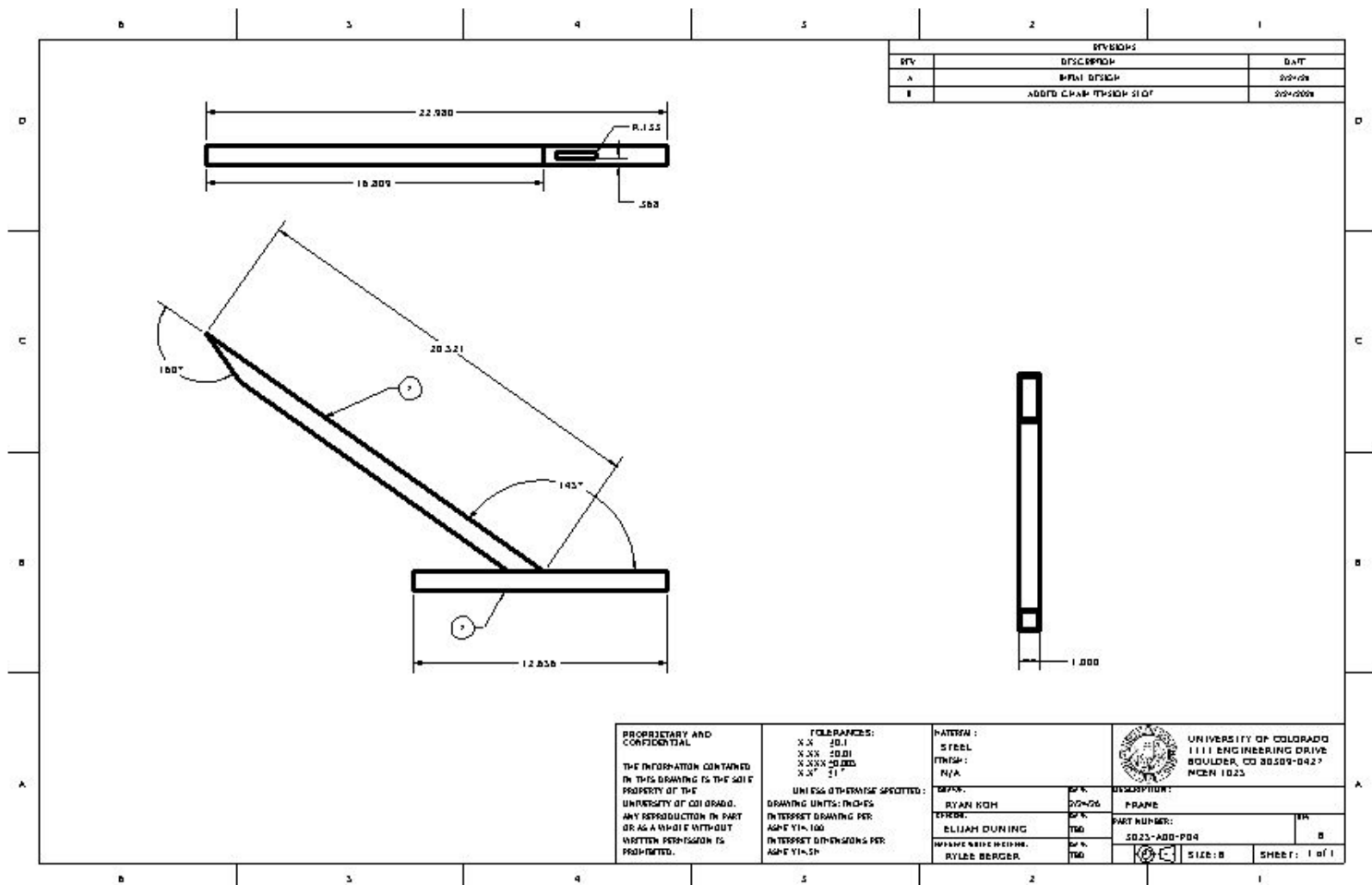
Frame Overview



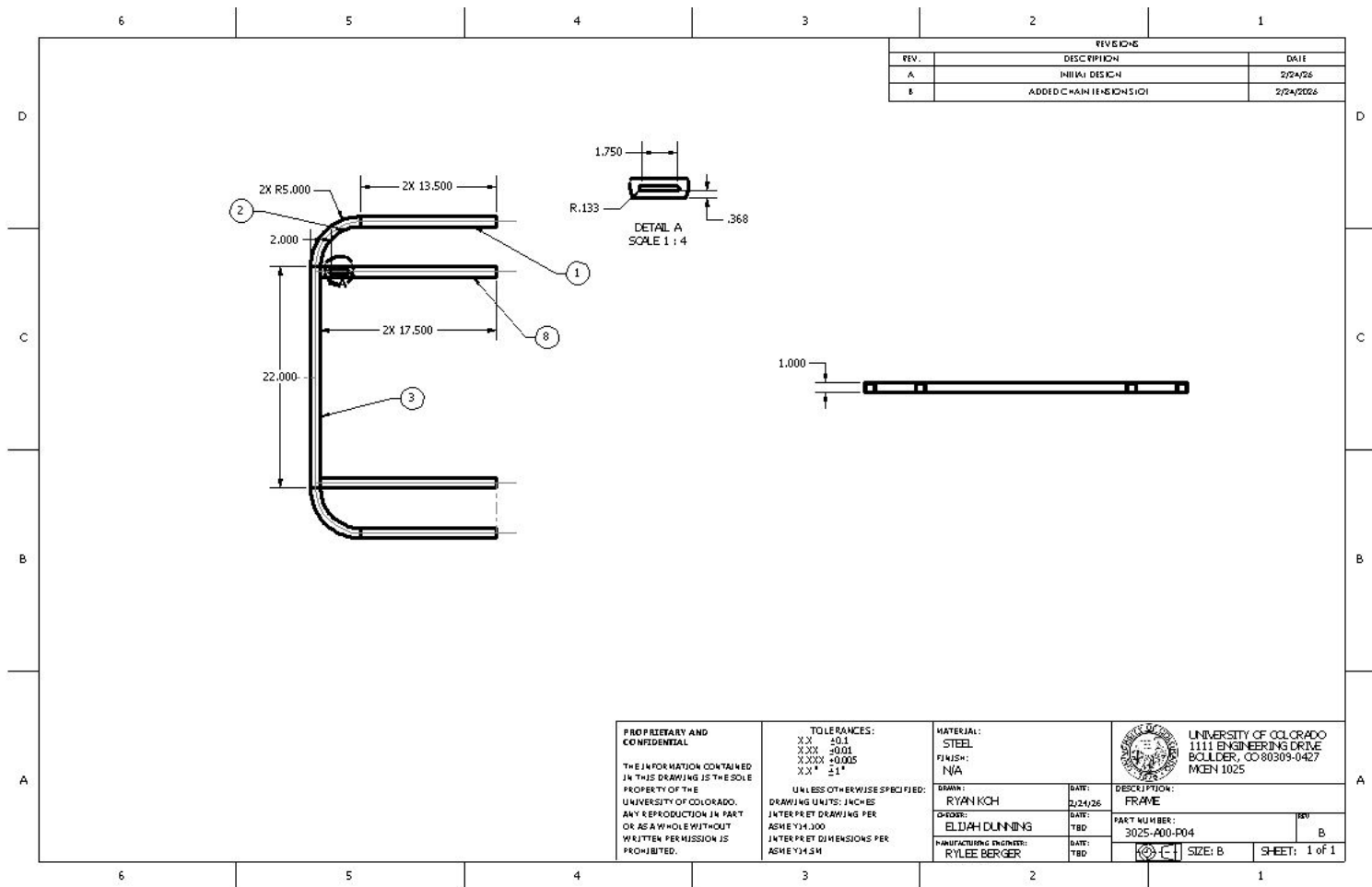
Weldments



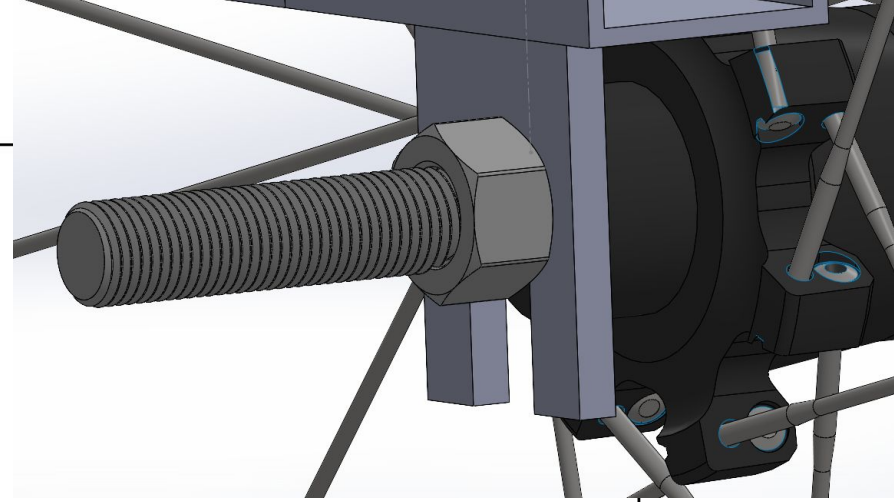
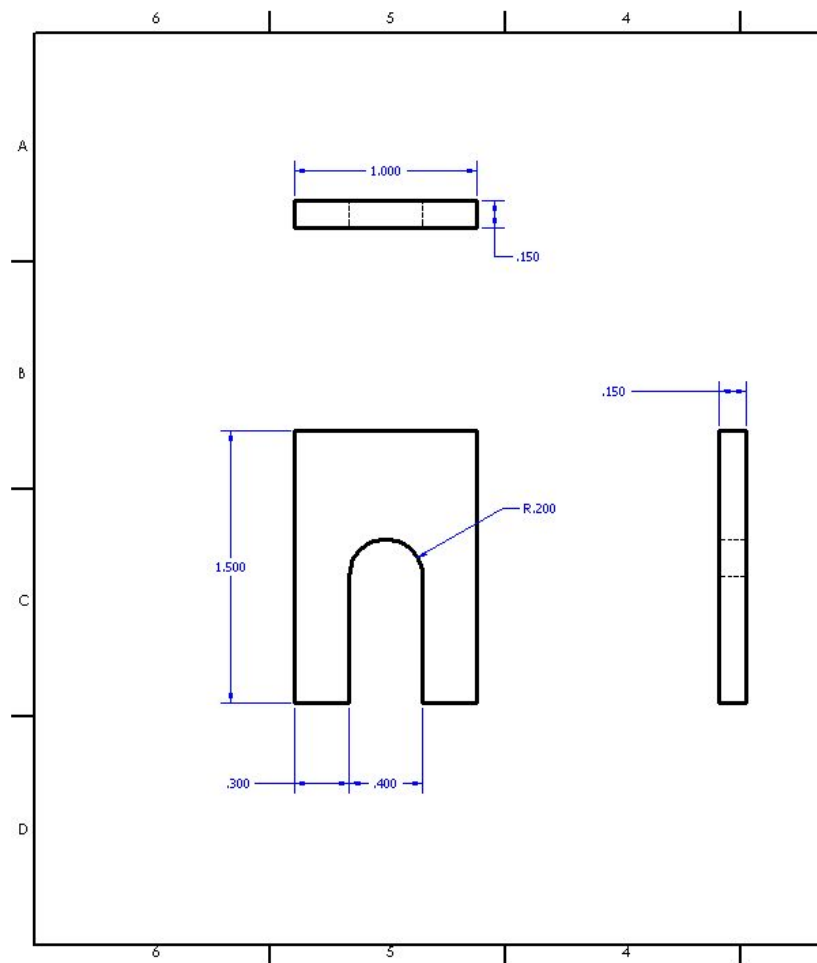
Weldments



Weldments



Wheel Mounts (Drop-out Wheels)



SPONSOR		dc DESIGN CENTER COLORADO	
PROJECT		FILE NAME	
TEAM NUMBER		wheel slot	
TOLERANCES	X.X ±0.1	APPROVAL	
UNLESS	X.XX ±0.01		
NOTED:	X.XXX ±0.005		
UNITS: INCH=ES	X.X" ±"		
DRAWN BY	ELIAH	DATE	2/24/2026
DATE		DESCRIPTION	
STOCK SIZE		PN	REV
		A	1
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Drill Plate Assembly

6

5

4

3

2

1

A

B

C

D

REV.	DESCRIPTION	DATE
ITEM NO.	PART NUMBER	QTY.
1	Drill Plate	1
2	Pillow Block	2
3	6383K39	2
4	8880T92	2
5	Pillow Block Shaft	1
6	2500T411	1
7	91251A627	7

SPONSOR

PROJECT

TEAM NUMBER

ITEM NAME

Drill Plate SubAssembly

DESIGNED BY

DATE

DESCRIPTION

APPROVED

DATE

REV

1 of 1

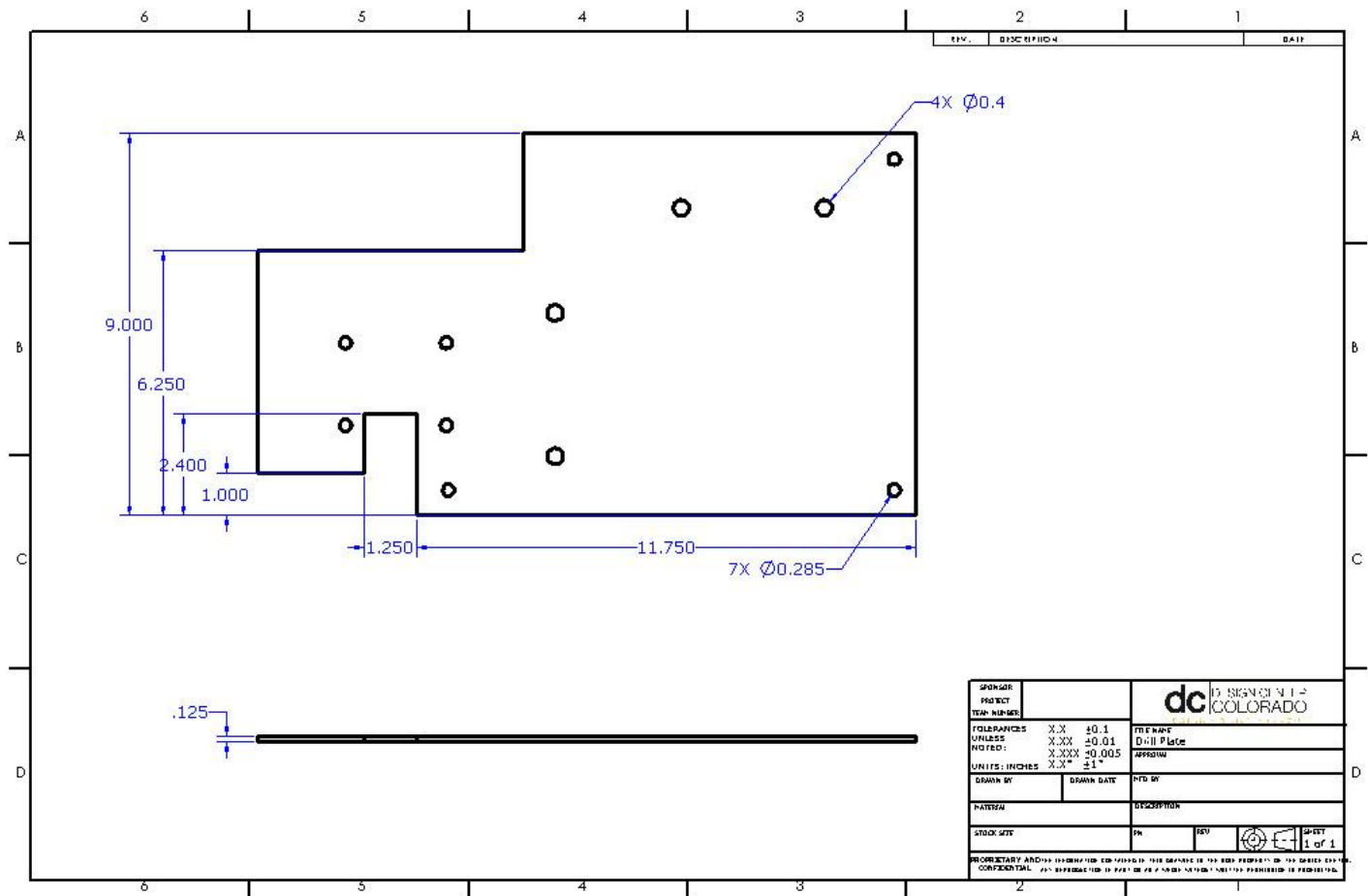
dc

DESIGN CENTER

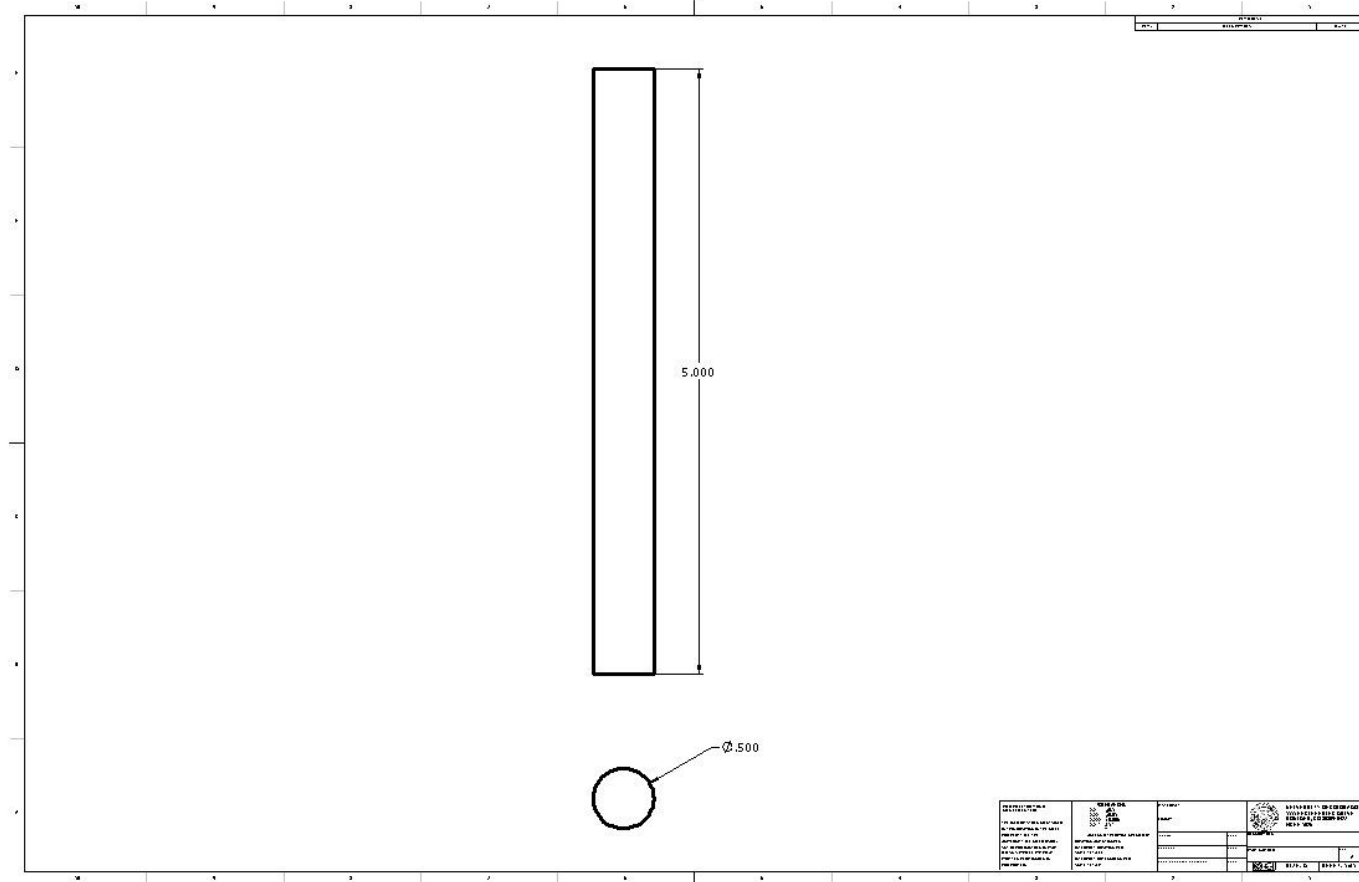
COLORADO

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Drill Plate



Pillow Block Shaft



Technical drawing of a mechanical part, likely a bracket or base plate, showing dimensions and a detail view of a threaded hole.

Dimensions:

- Overall width: 2.500
- Overall height: 2.750
- Distance from top edge to center of hole: 2.500
- Distance from left edge to center of hole: .133
- Radius of top corners: R.250
- Hole diameter: $\varnothing 1.281$
- Distance from right edge to center of hole: 2X 1.500

Detail View:

- Thread specification: 2X 1/4-20 UNC-2A
- Distance from left edge to center of hole: .625

Table:

REV.	DESCRIPTION	DATE

SPONSOR: PROJECT TEAM NUMBER

TOLERANCES: X.X ± 0.1
UNLESS NOTED: X.XX ± 0.01
 X.XXX ± 0.005

UNITS: INCHES X.XX $\pm .1^\circ$

DESIGN CENTER COLORADO

REFERENCE: Pillow Block

APPROVAL:

DRAWN BY: **DRAWN DATE:** **PTD BY:**

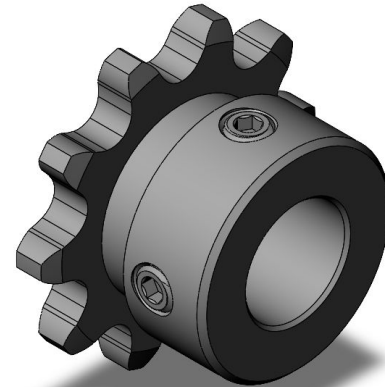
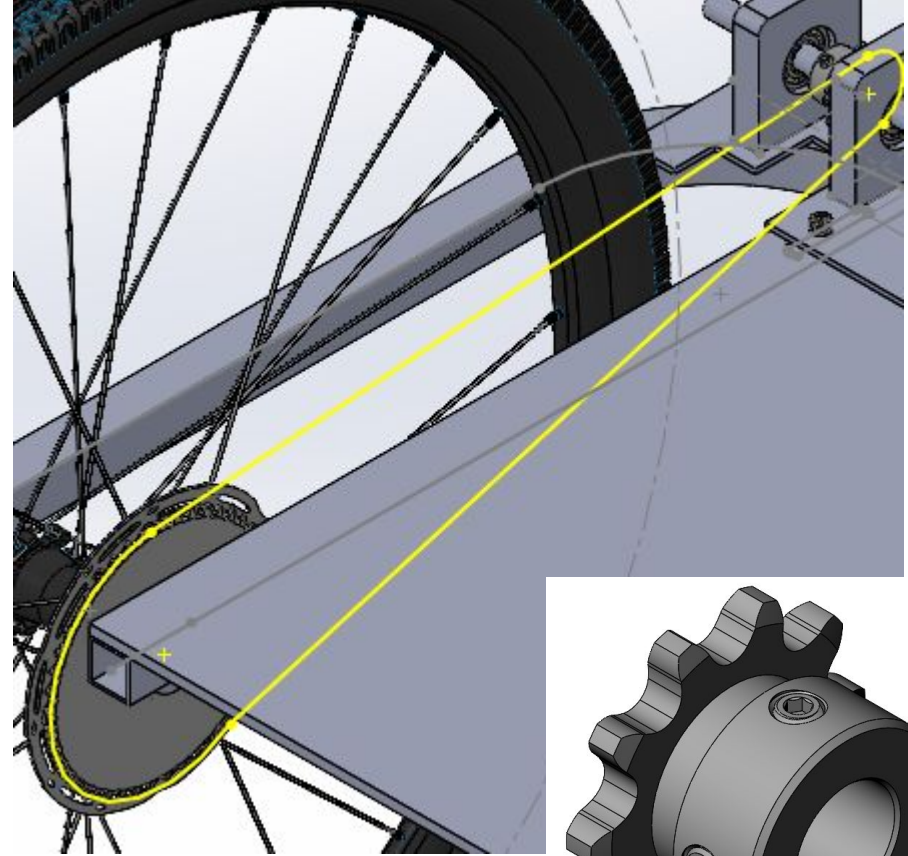
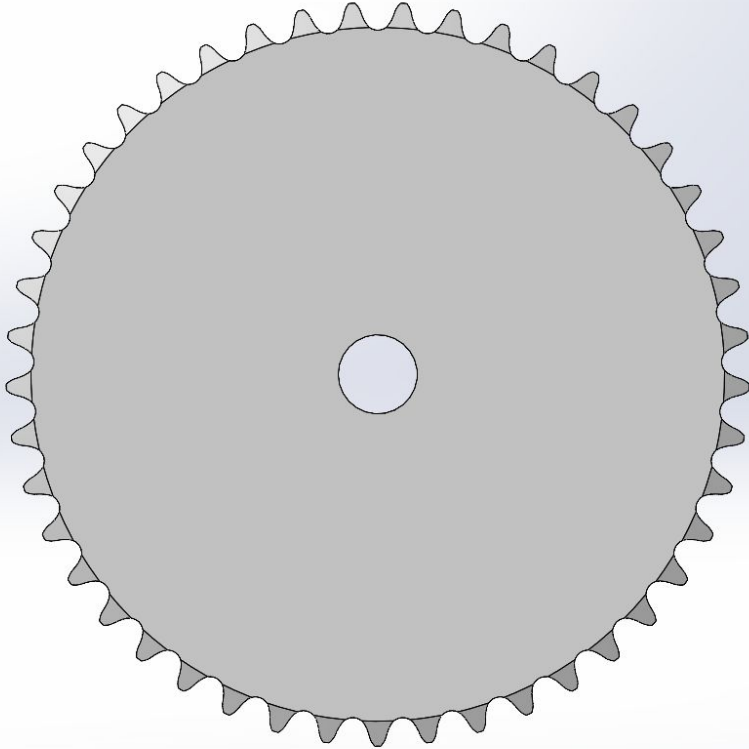
PATTERN: **DESCRIPTION:**

STOCK SIZE: IN **REV:** **SHEET:** 1 of 1

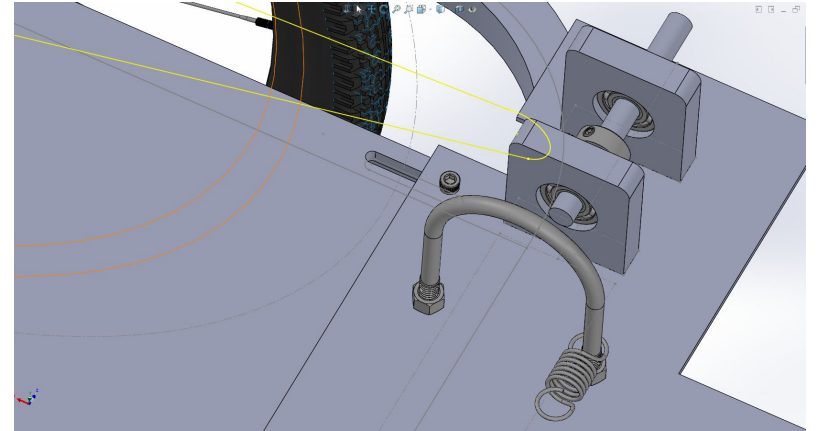
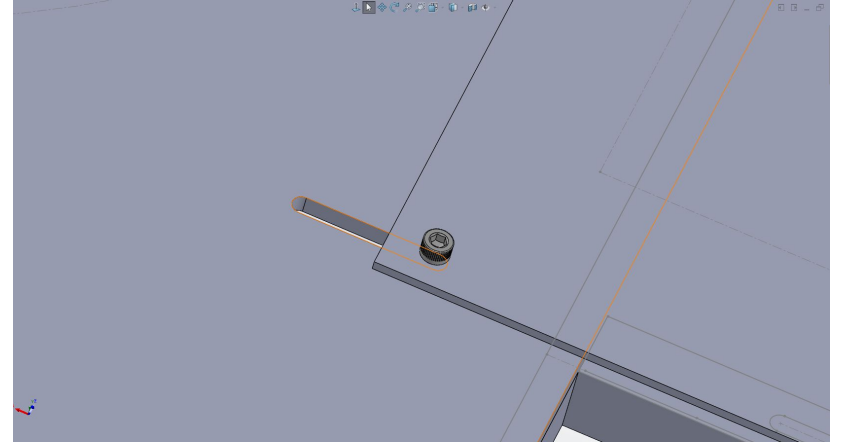
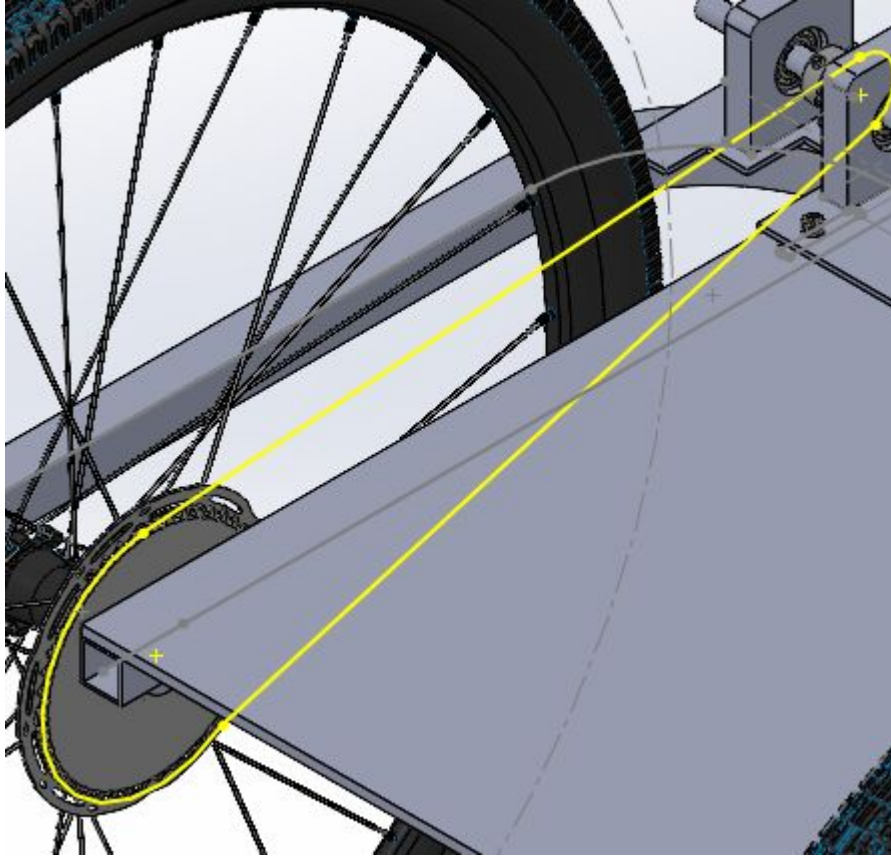
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Gears, Gear Train, Gear Ratio

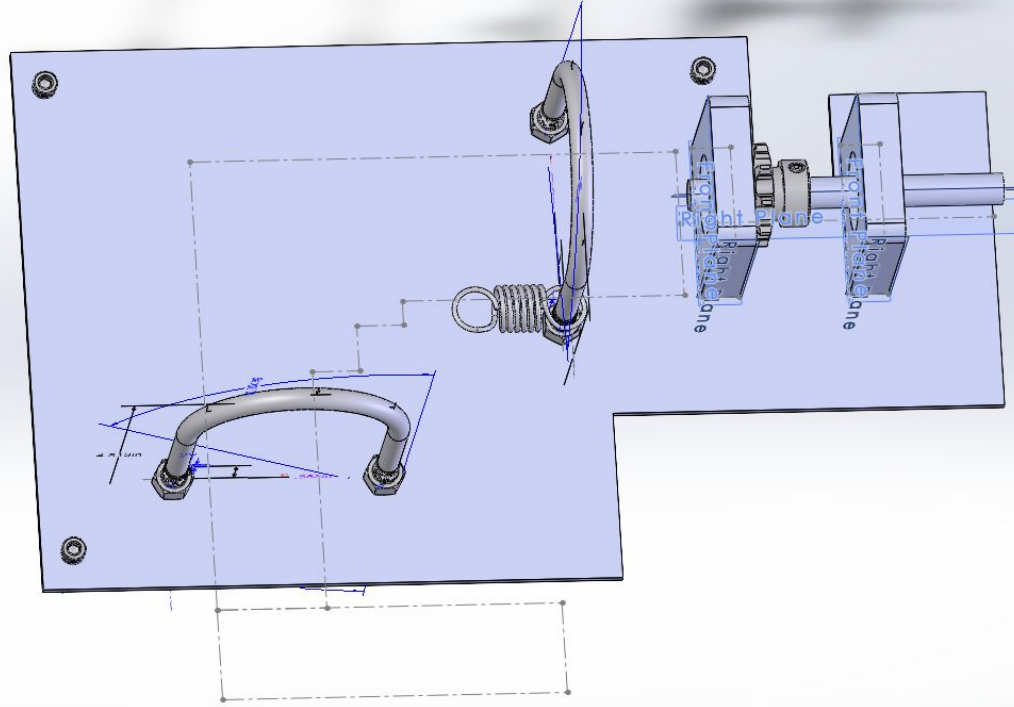
$$\text{Gear Ratio} = 45/10 = 4.5$$



Chain Line and Tension Plan



Actuation of Drill



Brake Design & Location



Calculations

Calculations

Total assembly weight: approx. 46lbs (40.8977lbs from SolidWorks + Fastener weight + Drill Weight + Error in this calculation)

Gear Ratio: 4.5:1

Torque: 271 lb in

RPM: 430rpm

Bill of Materials

Manufactured BOM

Part Number	Item Description	Quantity	Stock Material	Material Cost	Material Vendor	Responsible Drafter	Responsible Engineer	Received?
3025-A00-P01	Drill Plate Material	1	Aluminum	\$0.00	CU Boulder	N/A	Leighanne Bennett	In Progress
3025-A00-P02	Pillow Block Material	1	Aluminum	\$0.00	CU Boulder	N/A	Leighanne Bennett	In Progress
3025-A00-BP03	Drill Shaft Material	1	1566 ground shaft	\$0.00	CU Boulder	N/A	Leighanne Bennett	In Progress
3025-A00-P04	Sprocket Adapter and/or dropouts	As needed	Mild Steel	\$0.00	CU Boulder	N/A	Rylee Berger	In Progress
3025-A00-P05	Head Tube	1	TBD	\$0.00	CU Boulder	N/A	Rylee Berger	In Progress

COTS BOM

Part Number	Description	Quantity	Unit Cost	Total Cost	Vendor	Responsible Engineer	Received?
3025-A00-P01	Brakes	1	\$0.00	\$0.00	Ryan Koh	Ryan Koh	Yes
3025-A00-P02	Fasteners	TBD	TBD	TBD	McMaster-Carr	Rylee Berger	In progress
3025-A00-P03	Handle bars	1	\$0.00	\$0.00	Ryan Koh	Ryan Koh	Yes
3025-A00-P04	Front Frame	1	\$0.00	\$0.00	Ryan Koh	Ryan Koh	Yes
3025-A00-P05	Steel Stock, Square	TBD	TBD	TBD	TBD	Karl Kubala	In progress
3025-A00-P06	Bike Wheels 26"	3	TBD	TBD	TBD	Elijah Dunning	In progress
3025-A00-P07	1/4-20 UNC-2A (91251A527)	1 (25pk)	\$0.91	\$22.65	McMaster-Carr	Erik Stern	In progress
3025-A00-P08	10 Tooth Machine Bore Sprocket (6280K322)	1	\$17.92	\$17.92	McMaster-Carr	Erik Stern	In progress
3025-A00-P09	High Strength Wide Flange Nuts (1/4) (91627A115)	1	\$0.66	\$6.56	McMaster-Carr	Erik Stern	In progress
3025-A00-P10	Steel Extension Spring (5108N56)	1 (5pk)	\$3.53	\$17.65	McMaster-Carr	Erik Stern	In progress
3025-A00-P11	45 Tooth Flat Sprocket (2299K319)	1	\$33.62	\$33.62	McMaster-Carr	Erik Stern	In progress
3025-A00-P12	Steel 1/2" Shaft Ball Bearings (6383K39)	2	\$11.85	\$23.70	McMaster-Carr	Erik Stern	In progress
3025-A00-P13	3/8-16 Steel U Bolt ID 3" (8880T92)	4	\$3.48	\$13.92	McMaster-Carr	Erik Stern	In progress

Timeline

23-Feb-26							2-Mar-26							9-Mar-26							16-Mar-26							23-Mar-26						
Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
			Design Review & Revisions				Manufacturing I							SPRING BREAK							Manufacturing II													
			Order Parts				R E V I S I O N S	Order Parts				Begin Frame									Finish M-1 Components													
											Begin Drill Plate Components										Begin/Finish Drill Plate													
								→			Pillow Blocks										→ Bearings													
								→			Drill Shaft																							
											Purchased:																							
								→			Sprockets																							
											→ Machine Bore Sprocket																							
								→			Fasteners																							
30-Mar-26							6-Apr-26							13-Apr-26							20-Apr-26							27-Apr-26						
Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Assembly & Testing							Hardware Demonstration				Runoff Preparation & Competition Day														Final Report & Checkout									
Wheels			Testing				Draft Work				R E P O R T D R A F T	Video Recording				V E H I C L E D E M O	Runoff Preparation				R U N O F F D A Y	Report Work & Clean-up						R E P O R T D U E						
Brakes																																		